论文分类打榜赛

书生大模型实战营（沐曦魔乐专场）—— 论文分类打榜赛

在参赛前请先查看： 闯关手册——书生大模型实战营（沐曦+魔乐）专场

温馨提示：一定要记得关开发机，一定要记得关开发机，一定要记得关开发机，不然钱会用完！！！

1. 背景

在人工智能大语言模型蓬勃发展的时代背景下，书生·浦语（InternLM）凭借其卓越的知识理解、文本分析、数学推理和多语言处理能力，在各类评测任务中表现优异。该模型在综合性考试中成绩突出，在多项中文基准测试中超越了同期主流模型的表现。随着版本的持续迭代，InternLM在数理计算、代码生成、对话交互、内容创作等领域均实现了显著提升，整体性能位居同参数量级开源模型的前列。

本次书生·浦语端侧小模型**（1.8B）**论文分类微调竞赛旨在深入探索InternLM在垂直领域应用的巨大潜力。竞赛依托魔乐社区（Modelers）平台举办，魔乐社区是天翼云与华为联合推出的AI开发者社区，提供TDMA（工具链、数据集、模型、应用）的托管展示服务和支撑系统。魔乐社区汇聚AI产业链资源，依托理事会成员单位，吸引开发者共同推动AI发展，解决行业难题，促进生态繁荣。社区提供免费算力，方便用户体验AI模型和应用效果。魔乐社区的核心功能涵盖模型托管与管理、数据集托管、体验空间、Git仓库服务，以及openMind Library和openMind Hub Client等工具套件集成，为模型开发和管理提供了完善的技术支撑。

在算力支持方面，本次竞赛得到了沐曦的强力支撑。沐曦致力于为异构计算提供全栈GPU芯片及解决方案，可广泛应用于智算、智慧城市、云计算、自动驾驶、数字孪生、元宇宙等前沿领域，为数字经济发展提供强大的算力支撑。沐曦先进的GPU计算架构为参赛者提供了高效稳定的训练环境，确保模型微调过程的顺利进行。

竞赛采用学术资源丰富的arXiv数据集作为训练基础，该数据集汇聚了物理学、数学、计算机科学、定量生物学、定量金融等多个前沿学科的海量论文资源。arXiv作为全球领先的预印本论文发布平台，以其快速的论文更新频率和前沿的研究内容，为模型训练提供了高质量的学术数据支撑。同时，竞赛还整合了额外采集的学术论文数据，这些数据来源多样化，与原始arXiv数据集形成有效互补，进一步增强了训练数据的丰富性和覆盖面，有助于提升模型对不同研究风格和主题论文的精准分类能力。

竞赛设置了26个细分学科类别，具体映射关系如下：

量子物理（quant-ph）→ A，化学物理（physics.chem-ph）→ B，原子物理（physics.atom-ph）→ C，软凝聚态物理（cond-mat.soft）→ D，机器人学（cs.RO）→ E，计算语言学（cs.CL）→ F，软件工程（cs.SE）→ G，信息检索（cs.IR）→ H，高能理论物理（hep-th）→ I，高能现象学（hep-ph）→ J，光学（physics.optics）→ K，人工智能（cs.AI）→ L，计算机视觉（cs.CV）→ M，核理论（nucl-th）→ N，天体物理（astro-ph）→ O，概率论（math.PR）→ P，操作系统（cs.OS）→ Q，信号处理（eess.SP）→ R，最优化与控制（math.OC）→ S，动力系统（math.DS）→ T，微分几何（math.DG）→ U，数学物理（math.MP）→ V，多媒体（cs.MM）→ W，统计方法（stat.ME）→ X，组合数学（math.CO）→ Y，神经与进化计算（cs.NE）→ Z。

为确保竞赛的科学性和模型泛化能力的真实性，避免参赛者过度拟合训练数据而影响模型的实际应用价值，本次竞赛的测试集采用了20种不同的问题模板进行评估。现公开其中一种标准模板格式：

Code block

```
"Based on the title '{title}', authors '{authors}', and abstract '{summary}', please determine the scientific category of this paper."
```

在技术实施层面，所有参赛者必须基于**InternLM 1.8B**模型进行微调训练。参赛者需要根据竞赛提供的数据集特征，结合自身的技术储备和对模型架构的深度理解，并严格遵循相关的技术标准和学术伦理规范，确保最终模型在论文分类任务中具备良好的稳定性、准确性和可靠性。

2. 数据集介绍

https://www.kaggle.com/datasets/Cornell-University/arxiv

本比赛选取arXiv数据集作为核心数据来源，通过系统性数据分析确定了26个重要学术类别，包括量子物理（quant-ph）、化学物理（physics.chem-ph）、原子物理（physics.atom-ph）、软凝聚态物理（cond-mat.soft）、机器人学（cs.RO）、计算语言学（cs.CL）、软件工程（cs.SE）、信息检索（cs.IR）、高能理论物理（hep-th）、高能现象学（hep-ph）、光学（physics.optics）、人工智能（cs.AI）、计算机视觉（cs.CV）、核理论（nucl-th）、天体物理（astro-ph）、概率论（math.PR）、操作系统（cs.OS）、信号处理（eess.SP）、最优化与控制（math.OC）、动力系统（math.DS）、微分几何（math.DG）、数学物理（math.MP）、多媒体（cs.MM）、统计方法（stat.ME）、组合数学（math.CO）、神经与进化计算（cs.NE）。在此分类框架基础上，进一步追踪采集各对应类别的最新学术论文，通过科学的数据整合方法构建高质量的混合训练数据集。

所构建的数据集涵盖预训练数据和监督微调（SFT）数据两个层次，为InternLM模型的论文分类能力训练提供多维度的数据支撑。同时，为确保评估的科学性和可信度，验证集和测试集均采用20种不同问题模板进行构建，既保持了与训练数据的同源性，又通过模板多样化有效防止了过拟合现象，从而为参赛者的模型优化和竞赛评估奠定了坚实可靠的数据基础。

2.1 Arxiv dataset样例

Code block


```
{"id": "0711.3691",
 "submitter": "Eric Laporte",
 "authors": "Olivier Blanc (IGM-LabInfo), Matthieu Constant (IGM-LabInfo), Eric Laporte (IGM-LabInfo)",
 "title": "Outilex, plate-forme logicielle de traitement de textes \'ecrits",
 "comments": null,
 "journal-ref": "Dans Verbum ex machina. Proceedings of TALN - Outilex, plate-forme logicielle de traitement de textes \'ecrits, Louvain : Belgique (2006)",
 "doi": null,
 "report-no": null,
 "categories": "cs.CL",
 "license": null,
```

```
 "abstract": "The Outilex software platform, which will be made available to research, development and industry, comprises software components implementing all the fundamental operations of written text processing: processing without lexicons, exploitation of lexicons and grammars, language resource management. All data are structured in XML formats, and also in more compact formats, either readable or binary, whenever necessary; the required format converters are included in the platform; the grammar formats allow for combining statistical approaches with resource-based approaches. Manually constructed lexicons for French and English, originating from the LADL, and of substantial coverage, will be distributed with the platform under LGPL-LR license.",
 "versions": [
 {"version": "v1", "created": "Fri, 23 Nov 2007 09:45:13 GMT"},
 {"version": "v2", "created": "Tue, 27 Nov 2007 10:22:14 GMT"}
 ],
 "update_date": "2007-11-27",
 "authors_parsed": [
 ["Blanc", "Olivier", "", "IGM-LabInfo"],
 ["Constant", "Matthieu", "", "IGM-LabInfo"],
 ["Laporte", "Eric", "", "IGM-LabInfo"]
 ]}
```


2.2 抽取数据集样例

Code block


```
{"system": "你是个优秀的论文分类师",
"conversation":
[{"human": "Based on the title 'Theory of linear sweep voltammetry with diffuse charge: unsupported electrolytes, thin films, and leaky membranes', authors 'David Yan, Martin Z. Bazant, P. M. Biesheuvel, Mary C. Pugh, and\n  Francis P. Dawson', and abstract 'Linear sweep and cyclic voltammetry techniques are important tools for electrochemists and have a variety of applications in engineering. Voltammetry has classically been treated with the Randles-Sevcik equation, which assumes an electroneutral supported electrolyte. In this paper, we provide a comprehensive mathematical theory of voltammetry in electrochemical cells with unsupported electrolytes and for other situations where diffuse charge effects play a role, and present analytical and simulated solutions of the time-dependent Poisson-Nernst-Planck (PNP) equations with generalized Frumkin-Butler-Volmer (FBV) boundary conditions for a 1:1 electrolyte and a simple reaction. Using these solutions, we construct theoretical and simulated currentvoltage curves for liquid and solid thin films, membranes with fixed background charge, and cells with blocking electrodes. The full range of dimensionless parameters is considered, including the dimensionless Debye screening length (scaled to the electrode separation), Damkohler number (ratio of characteristic diffusion and reaction times) and dimensionless sweep rate (scaled to the thermal voltage per diffusion time). The analysis focuses on the coupling of Faradaic reactions and diffuse charge dynamics, although capacitive charging of the electrical double layers (EDL) is also studied, for early time transients at reactive electrodes and for non-reactive blocking electrodes. Our work highlights cases where diffuse charge effects are important in the context of voltammetry, and illustrates which regimes can be approximated using simple analytical expressions and which require more careful consideration.', please determine the scientific category of this paper.\n\nA. quant-ph\nB. physics.chem-ph\nC. physics.atom-ph\nD. cond-mat.soft\nE. cs.RO\nF. cs.CL\nG. cs.SE\nH. cs.IR\nI. hep-th\nJ. hep-ph\nK. physics.optics\nL. cs.AI\nM. cs.CV\nN. nucl-th\nO. astro-ph\nP. math.PR\nQ. cs.OS\nR. eess.SP\nS. math.OC\nT. math.DS\nU. math.DG\nV. math.MP\nW. cs.MM\nX. stat.ME\nY. math.CO\nZ. cs.NE",
"assistant": "B"}]}
```


本次比赛将不限微调框架和微调数据，因此需要参赛选手自行对数据进行处理和优化，同时也应自行就行模型训练的调参等工作，后续的介绍中将使用xtuner 和 ms-swift 框架讲解如何微调。

2.3 评测集介绍

使用 opencompass 进行测评，测评集为单选，如论文存在多类别的情况下，能保证除对应选项外，其他类别均不在选项内。

arXiv > q-bio > arXiv:2506.15671

Quantitative Biology > Genomics

[Submitted on 18 Jun 2025]

Quantum-inspired algorithm for simulating viral response

D. O. Konina, D. I. Korbashov, I. V. Kovalchuk, A. A. Nizamieva, D. A. Chermoshentsev, A. K. Fedorov

Understanding the properties of biological systems is an exciting avenue for applying advanced approaches to solving corr
is optimization. In this work, we present the results of a proof-of-concept study that applies a quantum-inspired optimiza
activity in host responses. Reducing the problem to the Ising form allows the use of available quantum and quantum-inspi
problem. Our study paves the way for exploring the full potential of quantum and quantum-inspired optimization tools in

不在选项内

Comments: 35 pages, 6 figures

Subjects: **Genomics (q-bio.GN)**; Molecular Networks (q-bio.MN) Quantum Physics (quant-ph)

Cite as: arXiv:2506.15671 **[q-bio.GN]**

(or arXiv:2506.15671v1 **[q-bio.GN]** for this version)

https://doi.org/10.48550/arXiv.2506.15671

选项 A

Submission history

From: Daria Konina [view email]

[v1] Wed, 18 Jun 2025 17:51:21 UTC (834 KB)

这道题选 A

× 退出 newformat_sft_test_data.csv 782.86 KB 下载

question	A	B	C	D	E	F	G	H	I	J	answer
Based on the title 'The Quantum Primordial Black Holes, Dimensionless Small Parameter, Inflationary Cosmology and Non-Gaussianity', authors 'Alexander Shalyt-Margolin', and abstract 'In the present work consideration is given to the primordial black holes ({\bf pbhs}) in the Schwarzschild-de Sitter Metric with small mass (ultralight) in the preinflationary epoch. Within the scope of natural assumptions, it has been shown that the quantum-gravitational corrections ({\bf qgcs}) to the characteristics of such black holes can contribute to all the cosmological parameters, shifting them compared with the semiclassical consideration. These contributions are determined by a series expansion in terms of a small parameter dependent on the hole mass (radius). For this pattern different cases have been considered (stationary, black hole evaporation...). It has been demonstrated that involvement of ({\bf qgcs}) leads to a higher probability for the occurrence of such ({\bf pbhs}. Besides, high-energy deformations of Friedmann Equations created on the basis of these corrections have been derived for different patterns. In the last section of this work it is introduced a study into the contributions generated by the above-mentioned ({\bf qgcs}) in inflationary cosmological perturbations. Besides, it has been shown that non-Gaussianity of these perturbations is higher as compared to the semi-classical pattern.', please determine the scientific category of this paper. Additional info: 35 pages, Latex	astro-ph	cond-mat.mes-hall	cond-mat.mtrl-sci	cs.CL	cs.CV	cs.LG	gr-qc	hep-ph	hep-th	quant-ph	I
Based on the title 'Existence of travelling waves in discrete sine-Gordon rings', authors 'Guy Katriel', and abstract 'We prove existence results for travelling waves in discrete, damped, dc-driven sine-Gordon equations with periodic boundary conditions.', please determine the scientific category of this paper.	astro-ph	cond-mat.mes-hall	cond-mat.mtrl-sci	cs.CL	cs.CV	cs.LG	gr-qc	hep-ph	hep-th	quant-ph	C
Based on the title 'Linear and nonlinear dynamics of matter wave packets in periodic potentials', authors 'Th. Anker, M. Albiez, B. Eiermann, M. Taglieber, M. K. Oberthaler', and abstract 'We investigate experimentally and theoretically the nonlinear propagation of 87Rb Bose Einstein condensates in a trap with cylindrical symmetry. An additional weak periodic potential which encloses an angle with the symmetry axis of the waveguide is applied. The observed complex wave packet dynamics results from the coupling of transverse and longitudinal motion. We show that the experimental observations can be understood applying the concept of effective mass, which also allows to model numerically the three dimensional problem with a one dimensional equation. Within this framework the observed slowly spreading wave packets are a consequence of the continuous change of dispersion. The observed splitting of wave packets is very well described by the developed model and results from the nonlinear effect of transient solitonic propagation.', please determine the scientific category of this paper. Additional info: 7 pages, 4 figures	astro-ph	cond-mat.mes-hall	cond-mat.mtrl-sci	cs.CL	cs.CV	cs.LG	gr-qc	hep-ph	hep-th	quant-ph	C
Based on the title 'Collapsing Catalytic Classes', authors 'Michal Koucký, Ian Mertz, Edward Pyne, Sasha Sami', and abstract 'A catalytic machine is a space-bounded Turing machine with additional access to a second, much larger work tape, with the caveat that this tape is full, and its contents must be preserved by the computation. Catalytic machines were defined by Buhrman et al. (STOC 2014), who, alongside many follow-up works, exhibited the power of catalytic space ($CSPACE$) and in particular catalytic logspace machines (CL) beyond that of traditional space-bounded machines. Several variants of CL have been proposed, including non-deterministic and co-non-deterministic catalytic computation by Buhrman et al. (STACS 2016) and randomized catalytic computation by Datta et al. (CSR 2020). These and other works proposed several questions, such as catalytic analogues of the theorems of Savitch and Immerman and Szelepcs\'enyi. Catalytic computation was recently derandomized by Cook et al. (STOC 2025), but only in certain parameter regimes. We settle almost all questions regarding randomized and non-deterministic catalytic computation, by giving an optimal reduction from catalytic space with additional resources to the corresponding non-catalytic space classes. With regards to non-determinism, our main result is that \[CL=CNL\] and with regards to randomness we show \[CL=CPrL\] where $CPrL$ denotes randomized catalytic logspace where the accepting probability can be arbitrarily close to $1/2$. We also have a number of near-optimal partial results for non-deterministic and randomized catalytic computation with less catalytic space. We show catalytic versions of Savitch's theorem, Immerman-Szelepcs\'enyi, and the derandomization results of Nisan and Saks and Zhou, all of which are unconditional and hold for all parameter settings. Our results build on the compress-or-compute framework of Cook et al. (STOC 2025). Despite proving broader and stronger results, our framework is simpler and more modular.', please determine the scientific category of this paper.	astro-ph	cond-mat.mes-hall	cond-mat.mtrl-sci	cs.CL	cs.CV	cs.LG	gr-qc	hep-ph	hep-th	quant-ph	D
Based on the title 'Ghostly interactions in (1+1) dimensional classical field theory', authors 'Cédric Deffayet, Aaron Held, Shinji Mukohyama, Alexander Vikman', and abstract 'We investigate the classical stability of two coupled scalar fields with opposite-sign kinetic terms evolving in 1+1 dimensional Minkowski spacetime. In the first part, we characterise unquenched ghostly interactions and present numerical solutions that support the following statements. First, the classical instability is not instantaneous and can even be benign, i.e., free of finite-time singularities. Second, while the classical instability can cascade towards higher frequency excitations, it is not driven by high frequency modes: At fixed amplitude, high-frequency modes are more stable than low-frequency modes. In the second part, we demonstrate that the classical instability can be quenched by mass terms. In particular, we exemplify that heavy high-frequency ghost fields seem to not violate the decoupling theorem and can be integrated out classically. In the third part, we demonstrate how self-interactions can quench the instability, for instance, by postponing its onset to parametrically large times. Extrapolating numerical results at large but finite evolution time to infinite evolution time, we demonstrate that classical fluctuations around trivial and nontrivial field-theory vacua are increasingly long-lived with (i) smaller initial amplitude of fluctuations, (ii) higher initial frequency of fluctuations, (iii) larger masses of the fields, or (iv) weaker interaction coupling. Moreover, our numerical simulations for field-theoretical generalisations of some globally-stable ghostly mechanical models do not feature any instability.', please determine the scientific category of this paper. Additional info: 18 pages, 13 figures + 1 appendices and references; source code available on https://github.com/aaron-hd/ghostlyPDE_1D; animations available on https://zenodo.org/records/15209689	astro-ph	cond-mat.mes-hall	cond-mat.mtrl-sci	cs.CL	cs.CV	cs.LG	gr-qc	hep-ph	hep-th	quant-ph	I
Based on the title 'Fast detection and reconstruction of merging Massive Black Hole Binary signals', authors 'Senwen Deng, Stanislav Babak, Sylvain Marsat', and abstract 'The Laser Interferometer Space Antenna (LISA) will detect gravitational waves from the population of merging massive black holes binaries (MBHBs) throughout the Universe. The LISA data stream will feature many superposed signals from different astrophysical sources, requiring a global fit procedure. Most of the MBHB signals will be loud enough to be detected days or even weeks before the merger; and for those sources LISA will be able to predict the time of the merger well in advance of the coalescence, as well as an approximate position in the sky. In this paper, we present a fast detection and signal reconstruction scheme for massive black hole binaries in the LISA observation band. We propose: (i) a detection scheme for MBHB mergers allowing a first subtraction of these signals for the purpose of a global fit, and (ii) an efficient early detection scheme providing a time-of-merger estimate for a pre-merger signal, that will allow to trigger a protection period, placing LISA in ``do not disturb'' mode and enabling more detailed analysis that will facilitate multi-messenger observations. We highlight the effect of confusion of several overlapping in time MBHB signals in the pre-merger detection.',	astro-ph	cond-mat.mes-hall	cond-mat.mtrl-sci	cs.CL	cs.CV	cs.LG	gr-qc	hep-ph	hep-th	quant-ph	G

3. 比赛结果提交

模型上传之后千万不要换和删除

新模型上传到新的仓库

否则复查时出现问题当成绩无效处理

比赛权重测评提交：https://aicarrier.feishu.cn/share/base/form/shrcnCR6hPPtl3v3y2ZVXomhME9

比赛分数查询：https://aicarrier.feishu.cn/share/base/query/shrcn3UPdvR9e0D1REcjQ9LLNub

实时榜单：https://aicarrier.feishu.cn/share/base/dashboard/shrcnb7TtJErZiMVLDOM5wtHPJb

> 注意，最终的前60的B榜将会被重新测试，请不要更换权重和删除，避免出现无法验证结果的情况。

手把手详细细节请看方案一（沐曦 XTuner 微调 InternLM）（已完成）、方案二(沐曦 Swift 微调 InternLM)（正在更新中）

4. Baseline

4.1 Baseline 方案（一）沐曦 XTuner 微调 InternLM 赛事 （已完成）

沐曦 XTuner 微调 InternLM 赛事 Baseline

4.2 （还在更新中）Baseline 方案（二）沐曦 Swift 微调 InternLM 赛事（正在更新中）

5. 赛事激励

8 月 5 日 21:00 B 榜排行榜前 60 名（需要提供有效训练日志及复现文档、提交的模型仅能在沐曦算力上进行训练）共享 10万 奖金池。

注意：

- **不要一个人多个号提交，一经发现取消竞赛资格。**
- **组委会会对最终前 60 名的队伍的比赛方案及面试抽查，检查模型是否是选手本人训练出来的。**

论文分类打榜赛

(*奖金为个税前)

沐曦硬件微调 InternLM 论文分类打榜赛，8月5日

B榜前 60名 共享近 10万元 奖金池

第 1 名	¥6000
第 2 名	¥5000
第 3 名	¥4000
第 4 – 20 名	¥2000
第 21 – 60 名	¥1000